

**Maxillary
Premolars**

Dentition

Premolars

General characteristics

The premolars are so named because they are situated mesial to permanent molars.

They replace deciduous molars, so they are succedaneous teeth.

There are 2 premolars in each quadrant.

They have usually 2 cusps, buccal and lingual so they are called bicuspid, this name is misleading because the lower second premolar may have 3 cusps.

They are developed from **4 lobes** as anterior teeth, 3 buccal lobes and 1 lingual lobe, **except the lower second premolar may have 5 lobes.**

The middle buccal lobe of the premolars is highly developed, it corresponds to the middle labial lobe of the canine.

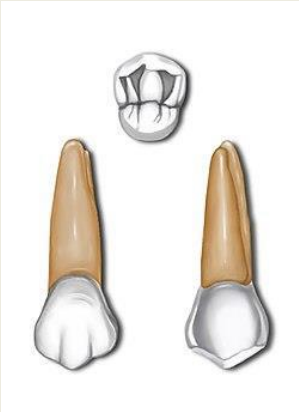
The cusps of upper and lower first premolars are more sharp than those of second premolars.

The premolars differ from anterior teeth in the following:-

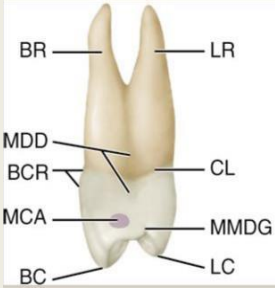
- a) They have occlusal surface with well-developed cusps instead of incisal edges.
- b) Their marginal ridge is part of their occlusal surface.
- c) Their crowns are shorter occluso-cervically.
- d) The contact areas of premolars (and molars) are nearly at the same level, broad and more cervically than those of anterior teeth.
- e) Their cervical line is less curved proximally.

Function of premolars:-

- They assist the canine in **tearing** of food by the sharp cusps of first premolars, and **assist molars** in mastication of food by the nearly equal cusp levels of second premolar.
- The premolars function with molars in maintaining the vertical dimension of the face.
- They support the corners of the mouth and cheeks to keep them from sagging.

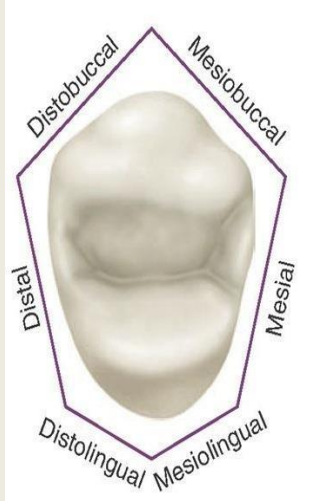
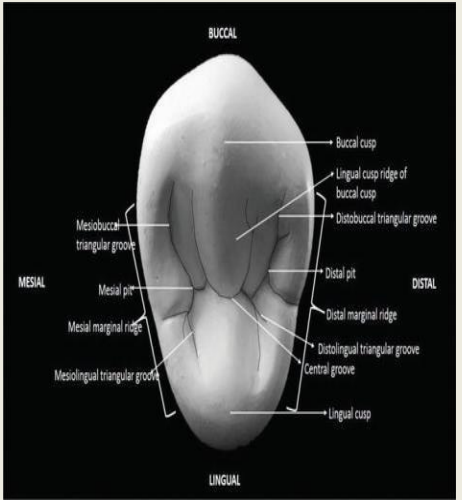
Buccal aspect	Upper first premolar	Upper second premolar
		
G.E.O.L	Trapezoid with the smallest uneven side cervically	
Outlines	Mesial outline: concave Distal outline: concave Cervical line: slightly curved rootwise Occlusal outline: mesial and distal slopes of the buccal cusp is concave . N.B mesial cusp slope is longer than distal cusp slope (This is opposite to the canine)	The same as upper 4 but :- 1-The crown is shorter. 2- The buccal cusp is less pointed and shorter. 3-the mesial cusp slope is smaller than the distal one.
MCA	In the middle third	In occlusal third
DCA	At the junction of the occlusal and middle thirds	More cervical
Elevation	1- Buccal ridge : extends occluso cervically in the middle. 2- Cervical ridge : at the cervical third.	All elevations are the same but less developed
Depressions	2 shallow vertical developmental grooves, mesial and distal to the buccal ridge.	
Root(s)	It may have 2 roots in 80 % of cases the buccal root is longer than the lingual root. The buccal root is curved distally.	Single root (higher percentage) and curved distally. 

Lingual aspect	Upper first premolar	Upper second premolar
		
The crown	The lingual surface is narrower than the buccal surface (lingual convergency)	
Outlines	Mesial outline: convex Distal outline: convex Cervical line: slightly curved rootwise The lingual cusp is shorter than the buccal cusp by 1 mm (to accommodate the tearing function of the first premolar rather than the grinding function in the second premolar. N.B distal cusp slope is longer than mesial cusp slope in lingual cusp.	The same as upper 4 but :- 1- the lingual cusp is nearly equal to the buccal cusp (to accommodate the grinding function of the 2nd premolar rather than the tearing function in the 1st premolar. 2- has less lingual convergency. To accommodate the shape of the dental arch .
Elevation	1- Buccal ridge : extends occluso cervically in the middle. 2- Cervical ridge : at the cervical third.	The lingual surface is highly convex with no lingual ridge.
Depressions	2 shallow vertical developmental grooves, mesial and distal to the buccal ridge.	
Root(s)	It may have 2 roots in 80 % of cases the buccal root is longer than the lingual root. The lingual root is curved distally.	Single root (higher percentage) and curved distally.

Mesial aspect	Upper first premolar	Upper second premolar
		
G.E.O.L	Trapezoid (مقلوب) with the smallest side occlusally.	
Outlines	Buccal outline: convex with the maximum convexity in the cervical 1/3 Lingual outline: convex with the maximum convexity is at the middle 1/3 cervical line :slightly curves crownwise. Occlusal outline: - Buccal cusp tip: is on the line with the buccal root apex - Buccal cusp is longer than lingual cusp by 1 mm. and the 2 cusp tips are sharp. - Lingual cusp tip: is on the line with the lingual border of the lingual root. - Mesial marginal ridge: in the middle 1/3	Buccal and lingual cusps are equal, this gives wider occlusal surface for grinding of food. MMR is located more occlusal.
Elevations And Depressions	-MMR crossed by mesial marginal developmental groove. -Mesial developmental depression (canine fossa) Located in the cervical third of crown and root trunk.	No Mesial marginal developmental groove No canine fossa slight mesial depression on the root.
Roots	-May be 2 roots or 1 root but majority is the bifurcated root at the middle third. -There is deep developmental depression along the root trunk which is called Canine fossa. -If one root , the root outline end in a blunt apex above the center of the crown.	Single root.

LM
LOGO MAKER

Distal aspect	Upper first premolar	Upper second premolar
outlines	Cervical line is less curved	
Elev. And Dep.	-No developmental groove crossin DMR. -No Canine fossa. -Distal contact area is broader and located more occlusaly. This is the opposite of all other posterior teeth. -DMR is occlusal than the mesial one, this isopposite of all other posterior teeth.	
Root	The bifurcation is more apically	Distal surface of the root has deeper developmental depression.

Occlusal aspect	Upper first premolar	Upper second premolar
		
Geol	Angular Hexagonal	Oval and less angular
Lingua l convergency	Severe	Less severe
Dimensions	Thickness > width	
Outlines	Mesiobuccal and distobuccal sides are equal in length. Mesiolingual side is shorter than the distolingual side. The mesial side is shorter than the distal side.	
Elevations	1) Buccal and lingual triangular ridges Over the buccal and lingual cusps. 2) MMR and DMR , the DMR is well developed .	The same but with thicker marginal ridges
Depressions	1) Mesial and distal triangular fossae. With pits. 2) Central developmental grooves, extends between the 2 fossae. This central developmental groove crosses the mesial marginal ridge as mesial marginal developmental groove.	The same but: -Shorter Central developmental groove. -Multiple supplemental grooves , so the Tooth appears wrinkled . -no mesial marginal developmental groove

Pulp cavities of upper premolars :

- Mesio-distally:** The pulp chamber is pointed occlusally.
The root canal is narrow.
- Bucco-lingually:** The pulp chamber has buccal and lingual pulp horns.
The upper 1st premolar mostly has **2 roots canals** even if it has only one root.
The upper 2nd premolar has mostly **one root canal**.

LM
LOGO MAKER